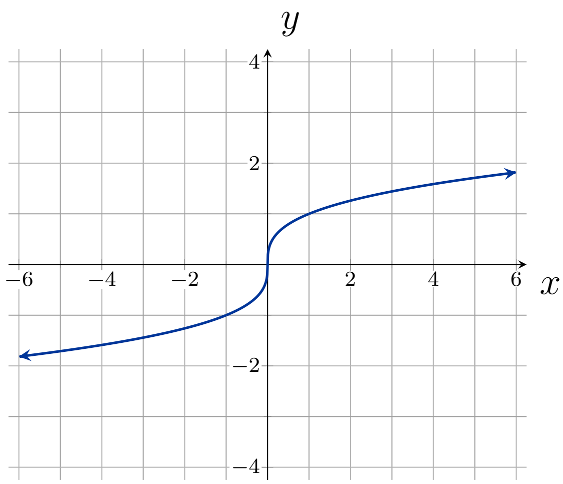
1.5: Transformation of Functions

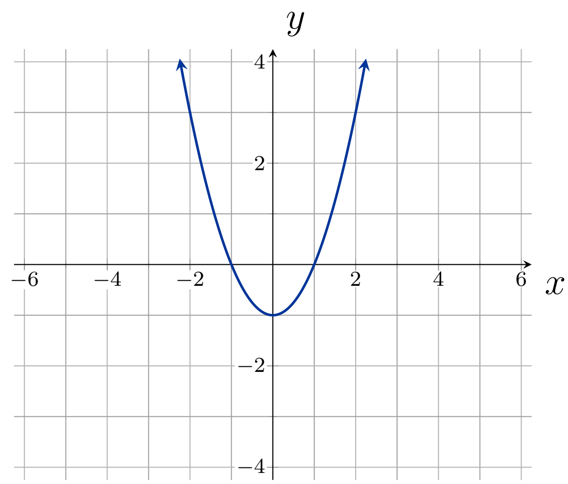
Definition. (Vertical Shift)
Given a function $f(x)$, a new function $g(x) = f(x) + k$, where k is a constant, is a **vertical shift** of the function $f(x)$. All the output values shift vertically by k units (**up** if $h > 0$, **down** if $h < 0$).

Example. Using the graphs below, sketch the shifted functions as indicated:

Graph $f(x) + 2$:



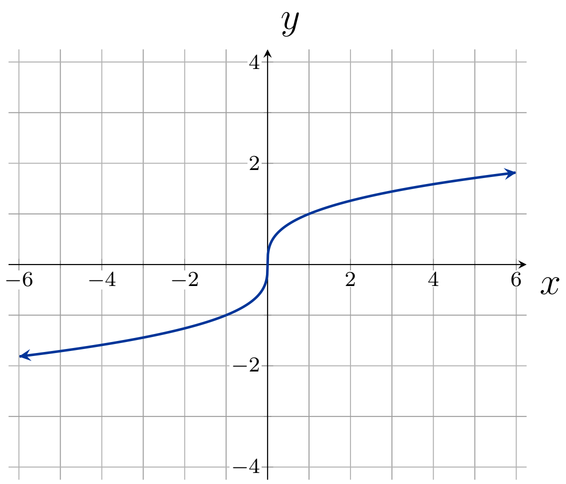
Graph $f(x) - 1$:



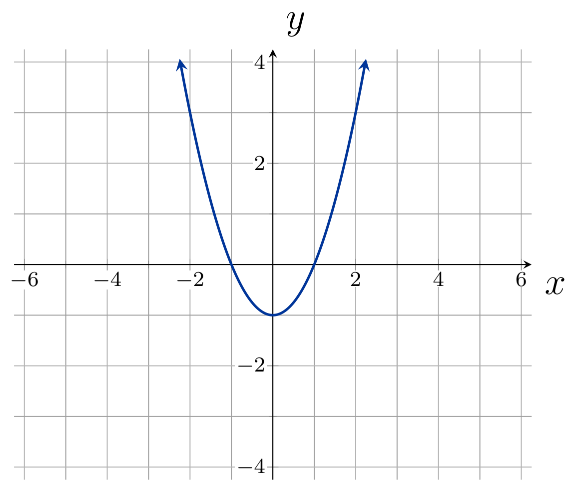
Definition. (Horizontal Shift)
Given a function $f(x)$, a new function $g(x) = f(x - h)$, where h is a constant, is a **horizontal shift** of the function $f(x)$. All the output values shift horizontally by h units (**right** if $k > 0$, **left** if $k < 0$).

Example. Using the graphs below, sketch the shifted functions as indicated:

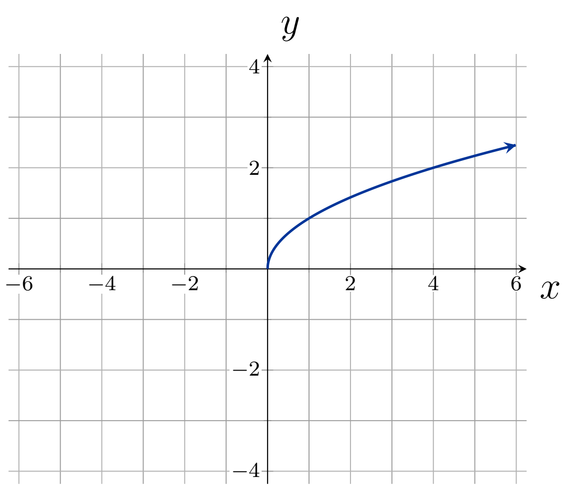
Graph $f(x + 2)$:



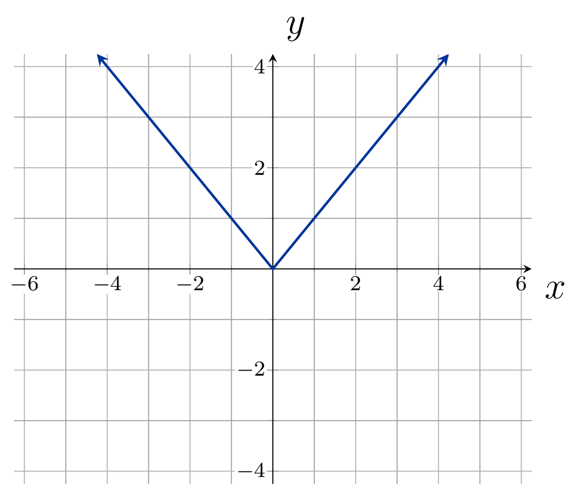
Graph $f(x - 1)$:



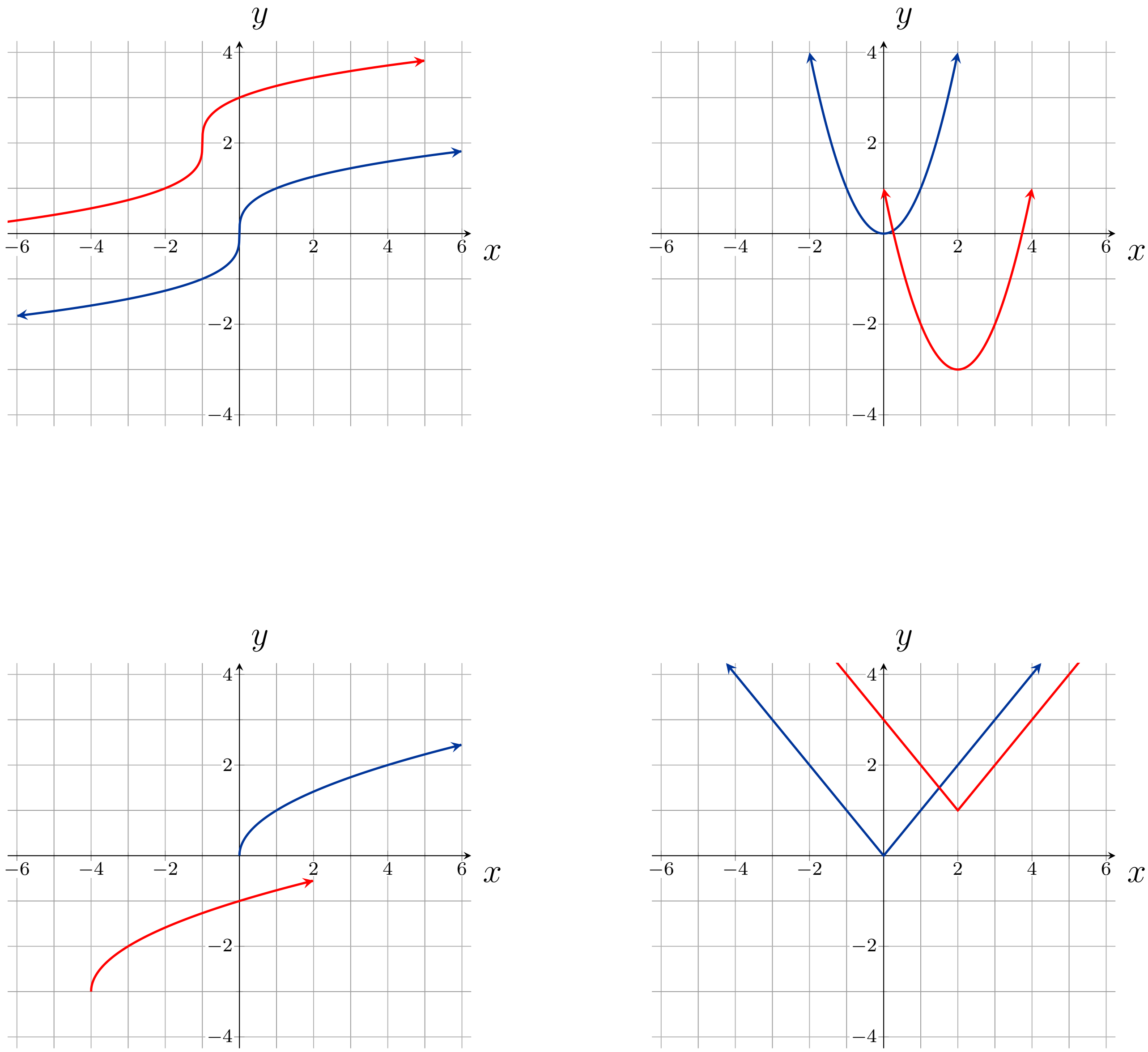
Graph $f(x - 2) + 1$:



Graph $f(x + 1) - 3$:



Example. Identify the shifts which transform the blue graph into the red graph:

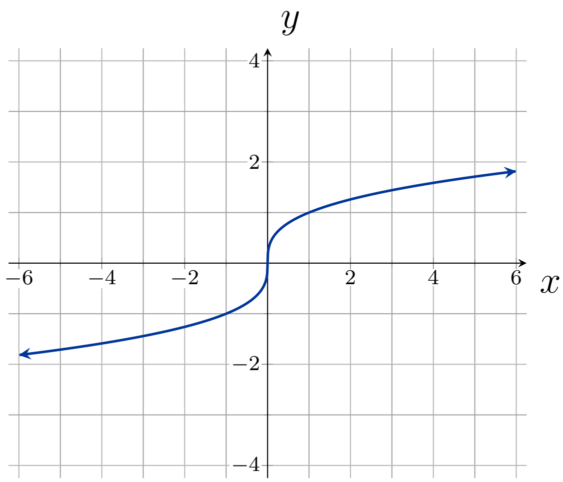


Definition. (Reflections)
Given a function $f(x)$, a new function $g(x) = -f(x)$ is a **vertical reflection** of the function $f(x)$, sometimes called a reflection about the x-axis.

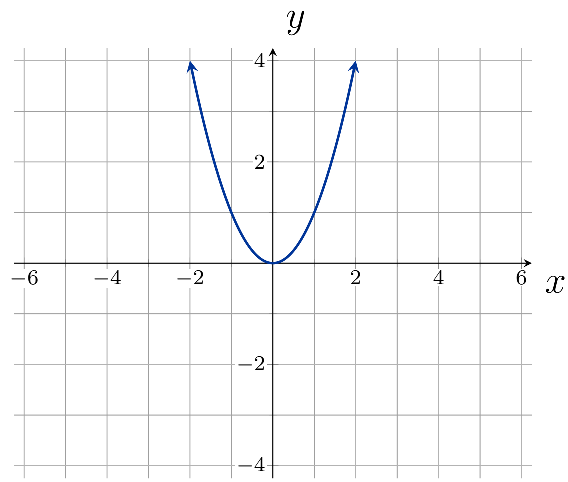
Given a function $f(x)$, a new function $g(x) = f(-x)$ is a **horizontal reflection** of the function $f(x)$, sometimes called a reflection about the y-axis.

Example. Transform the graphs below as indicated:

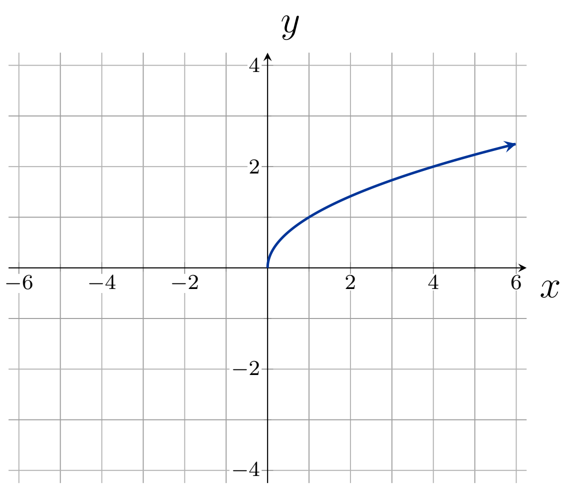
Graph $f(-x)$:



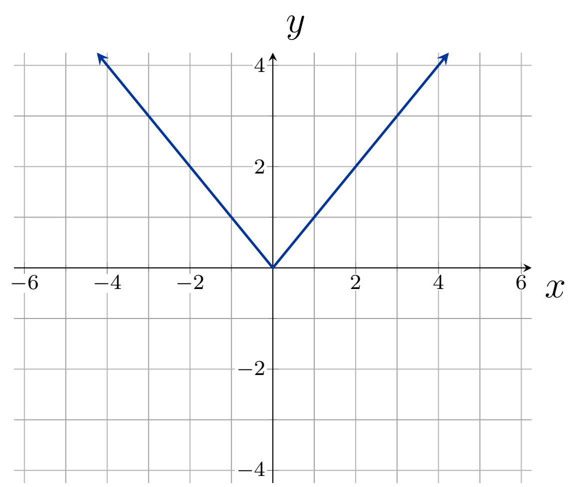
Graph $-f(x)$:



Graph $-f(-x)$:



Graph $-f(x) + 1$



Definition. (Even and Odd functions)
A function is called an even function if for every input x

$$f(x) = f(-x)$$

The graph of an even function is symmetric about the $y-$axis.

A function is called an odd function if for every input x

$$f(x) = -f(-x)$$

The graph of an odd function is symmetric about the origin.

Example. Using algebra, determine if the following functions are even, odd, or neither:
Graphs

$f(x) = x^3$

$g(x) = |x| + 1$

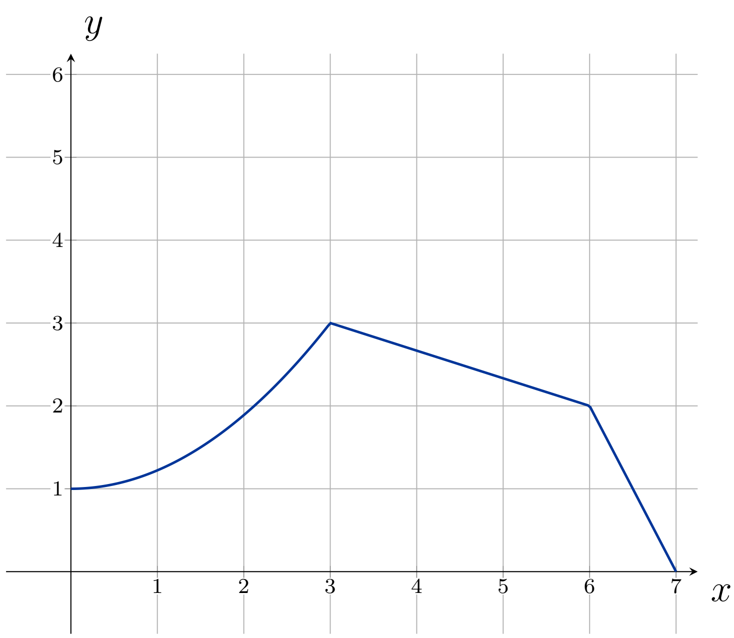
$h(x) = (x-1)^2$

$j(x) = x^3 - \dfrac{1}{x}$

Definition. (Vertical Stretches and Compressions)
Given a function $f(x)$, a new function $g(x) = af(x)$, where a is a constant, is a **vertical stretch** or **vertical compression** of the function $f(x)$.

- If $a > 1$, then the graph will be stretched.
- If $0 < a < 1$, then the graph will be compressed.
- If $a < 0$, then there will be a combination of a vertical stretch/compression with a vertical reflection.

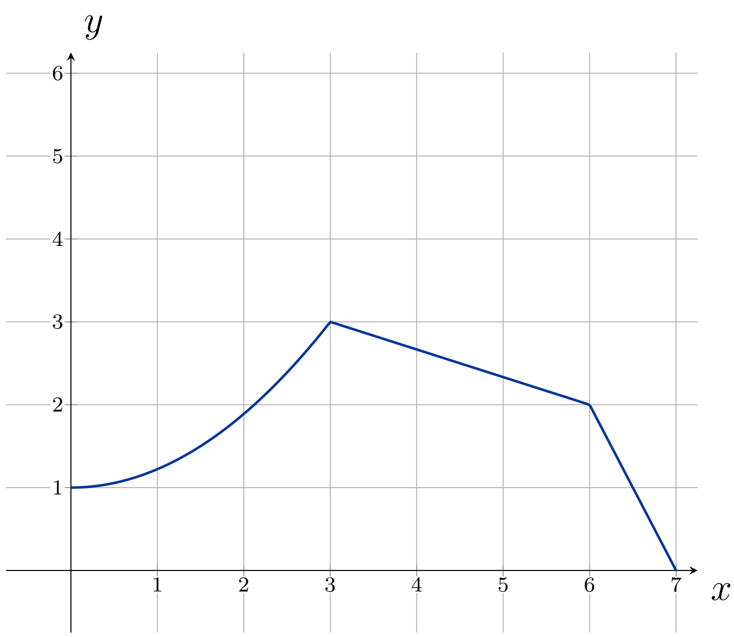
Example. Given the graph of $f(x)$ below, graph $g(x) = 2f(x)$: Graphs



Definition. (Horizontal Stretches and Compressions)
Given a function $f(x)$, a new function $g(x) = f(bx)$, where b is a constant, is a **horizontal stretch** or **horizontal compression** of the function $f(x)$.

- If $b > 1$, then the graph will be stretched by $\frac{1}{b}$.
- If $0 < b < 1$, then the graph will be compressed by $\frac{1}{b}$.
- If $b < 0$, then there will be a combination of a horizontal stretch/compression with a horizontal reflection.

Example. Given the graph of $f(x)$ below, graph $g(x) = f(2x)$: Graphs



Combining Transformations

When combining transformations ...

- $af(x) + k$: first vertically stretch by a and then shift by k.
- $f(bx - h)$: first horizontally shift by h and then horizontally stretch by $\dfrac{1}{b}$.
- $f(b(x - h))$: first horizontally stretch by $\dfrac{1}{b}$ and then horizontally shift by h.
- Horizontal and vertical transformations are independent of the order they are performed.

Example. Given the graph of $f(x)$ below, graph $g(x) = f\left(\dfrac{1}{2}x + 1\right) - 3$: Graphs